A Short Uniform Lower Bound for the Permanent

Why the Permanent is not in uniform TC^0 — and therefore not in uniform ACC^0

Abstract

We give a deliberately economical proof that the Permanent is not computable by DLOGTIME-uniform polynomial-size constant-depth threshold circuits. Since uniform ACC^0 is contained in uniform TC^0, this immediately also rules out uniform polynomial-size ACC^0 circuits for the Permanent. The proof is an indirect diagonalization due to Koiran and Perifel, specialized to constant threshold depth. It uses only three substantive facts: the Permanent is #P-complete under extremely local uniform reductions; if the Permanent were easy then PP = P; and under PP = P, a succinctly represented constant-depth threshold circuit can be evaluated in polynomial time. Scaling Valiant's local reduction from P to E would then imply $\mathrm{E} \subseteq \mathrm{P}$, contradicting the time hierarchy theorem. We conclude with the stronger Koiran–Perifel depth bound and the earlier Allender–Gore size bound.

1 The two statements

We consider the usual bit-complexity version of the Permanent on 0–1 matrices; equivalently, a circuit family computes the binary representation of $\mathrm{Perm}(M)$.

Theorem 1.1 (Koiran–Perifel, constant-depth specialization)**.** *The Permanent is not computable by a DLOGTIME-uniform family of polynomial-size constant-depth threshold circuits. Equivalently,*

$$\mathrm{Perm} \notin \mathrm{DLOGTIME}\text{-}\textit{uniform}\ \mathrm{TC}^0.$$

The ACC^0 consequence is immediate from the standard containment $\mathrm{ACC}^0 \subseteq \mathrm{TC}^0$.

Corollary 1.2. *The Permanent is not computable by DLOGTIME-uniform polynomial-size constant-depth* ACC^0 *circuits.*

The virtue of the proof below is that it does *not* use the Yao–Beigel–Tarui representation of ACC^0, Toda's theorem, or any direct lower-bound method for threshold circuits.

Proof in one line. Assume $\mathrm{Perm} \in$ DLOGTIME-uniform TC^0. Then PP = P. Local #P-completeness of the Permanent scales every language in E to a succinct exponentially large constant-depth threshold computation. Under PP = P, such a succinct threshold computation can be evaluated in polynomial time. Hence $\mathrm{E} \subseteq \mathrm{P}$, contradicting time hierarchy.

2 Ingredient I: easy Permanent collapses counting

Recall that a language is in PP if its membership can be decided by asking whether at least half of the computation paths of a polynomial-time nondeterministic machine accept. Also recall that the Permanent of 0–1 matrices is #P-complete.

Lemma 2.1. *If the Permanent has a P-uniform family of polynomial-size threshold circuits, then*

$$\mathrm{PP} = \mathrm{P}.$$

Proof. A polynomial-size threshold circuit can be evaluated in polynomial time: at a threshold gate, add the polynomially many input bits and compare the sum with the threshold. P-uniformity lets the algorithm recover the circuit itself. Thus the assumed family puts the Permanent in FP.

By #P-completeness of the Permanent, every #P function is then computable in polynomial time. In particular, every majority comparison of a #P count can be decided in polynomial time, so PP = P. □

DLOGTIME-uniformity is stronger than P-uniformity, so the hypothesis of the main theorem certainly gives Lemma 2.1.

3 Ingredient II: succinct threshold circuits become easy when PP = P

The key observation is best phrased for an exponentially large circuit that is given only by a short description.

Definition 3.1 (Succinct threshold circuit)**.** Let C be a threshold circuit of size s. Instead of listing C explicitly, we are given a polynomial-size Boolean circuit B such that $B(i)$ returns the ith bit of a standard encoding of C. We call B a *succinct description* of C.

If $s = 2^{O(n)}$, the gate names have only $O(n)$ bits, so a succinct description can still be manipulated in $\mathrm{poly}(n)$ time.

Lemma 3.2 (Constant-depth succinct evaluation)**.** *Assume* PP = P*. For every fixed constant d, the value of a depth-d threshold circuit of size s, given by a succinct description of size* $\mathrm{poly}(\log s)$*, can be computed in* $\mathrm{poly}(\log s)$ *time.*

Proof. Proceed by induction on the depth. At depth 0 there is nothing to do. Suppose values of gates of depth at most k can be computed in $\mathrm{poly}(\log s)$ time, and consider a threshold gate g one level higher.

The question is whether at least t_g of the at most s possible predecessors of g both

1. are actually wired into g, and
2. evaluate to 1.

Given the binary name of a candidate predecessor j, both conditions can be checked in $\mathrm{poly}(\log s)$ time: the first from the succinct circuit description, and the second by the inductive hypothesis.

Thus evaluating g is a threshold-counting question over $O(\log s)$ guessed bits, hence a PP computation on an input of length $\mathrm{poly}(\log s)$. Under PP = P, it can be answered in polynomial time. Since d is constant, iterating this argument through all d layers preserves $\mathrm{poly}(\log s)$ running time. □

Remark 3.3 (The quantitative Koiran–Perifel lemma). Koiran and Perifel track the induction uniformly in the depth and obtain the stronger running time

$$(\log s)^{2^{O(d)}}.$$

This is the only extra quantitative input needed to push the lower bound from constant depth all the way to depth $o(\log \log n)$.

4 Ingredient III: local completeness of the Permanent

For the scaling argument we need more than ordinary polynomial-time #P-hardness.

Proposition 4.1 (Local Valiant completeness)**.** *The Permanent of* 0–1 *matrices is* #P*-hard under DLOGTIME-uniform* AC^0 *many-one reductions. In particular, for every language* $A \in \mathrm{P}$ *there are DLOGTIME-uniform constant-depth polynomial-size circuits that, from* x*, produce a* 0–1 *matrix* $M(x)$ *and a simple postprocessing map* α *satisfying*

$$x \in A \quad \Longleftrightarrow \quad \alpha(\mathrm{Perm}(M(x))) = 1.$$

We will take Proposition 4.1 as the one imported completeness fact. It is a very strong uniform form of the usual Valiant reduction.

The point of such strong uniformity is that it scales cleanly.

Lemma 4.2 (Scaling from P to E)**.** *Suppose the Permanent has DLOGTIME-uniform polynomial-size constant-depth threshold circuits. Then every language in* E *has a family of constant-depth threshold circuits of size* $2^{O(n)}$ *whose individual wiring bits are computable in* $\mathrm{poly}(n)$ *time.*

Proof sketch. Let $L \in \mathrm{E}$, so some machine decides L in time 2^{cn} for a constant c. Pad an n-bit input to length $N = 2^{\Theta(n)}$. The padded computation is polynomial-time in N, so Proposition 4.1 reduces it to the Permanent of a matrix of dimension $N^{O(1)} = 2^{O(n)}$ using a constant-depth reduction whose wiring is locally computable.

Compose that reduction with the hypothesized DLOGTIME-uniform threshold circuits for Permanent. The resulting threshold circuit has size $2^{O(n)}$, constant depth, and a succinct description computable in polynomial time in n. □

5 The lower bound

We can now assemble the proof.

Proof of the main theorem. Assume for contradiction that the Permanent has DLOGTIME-uniform polynomial-size constant-depth threshold circuits.

By Lemma 2.1,

$$\mathrm{PP} = \mathrm{P}.$$

Take any $L \in \mathrm{E}$. By Lemma 4.2, on n-bit inputs L is computed by a constant-depth threshold circuit C_n of size

$$s = 2^{O(n)}$$

with a succinct description of size $\mathrm{poly}(n)$. By Lemma 3.2, because $\mathrm{PP} = \mathrm{P}$, the value of C_n can be computed in time

$$\mathrm{poly}(\log s) = \mathrm{poly}(n).$$

Therefore $L \in \mathrm{P}$. Since $L \in \mathrm{E}$ was arbitrary,

$$\mathrm{E} \subseteq \mathrm{P},$$

contradicting the deterministic time hierarchy theorem. Hence the Permanent is not in DLOGTIME-uniform polynomial-size TC^0. □

6 Why ACC^0 follows for free

For completeness, here is an explicit threshold simulation of a modular gate. Let

$$S = x_1 + \cdots + x_N.$$

For $j = 0, 1, \ldots, N$, form the threshold predicate

$$T_j = [S \geq j].$$

Then

$$[S = j] = T_j \wedge \neg T_{j+1}.$$

Hence for every fixed modulus m,

$$[S \equiv 0 \pmod m] = \bigvee_{\substack{0 \leq j \leq N \\ j \equiv 0 \pmod m}} [S = j].$$

Thresholds with arbitrary integer threshold can be implemented by majority gates using constants and padding, and the AND/OR layer can itself be realized by threshold gates. The construction is polynomial-size, constant-depth, and DLOGTIME-uniform. Therefore

$$\text{DLOGTIME-uniform } \mathrm{ACC}^0 \subseteq \text{DLOGTIME-uniform } \mathrm{TC}^0.$$

The ACC^0 lower bound follows immediately from the threshold lower bound.

7 The stronger theorem, and the historical picture

The argument above deliberately proves only the easiest constant-depth statement. Koiran and Perifel prove substantially more.

Theorem 7.1 (Koiran–Perifel, 2009)**.** *The Permanent has no DLOGTIME-uniform polynomial-size threshold circuits of depth*

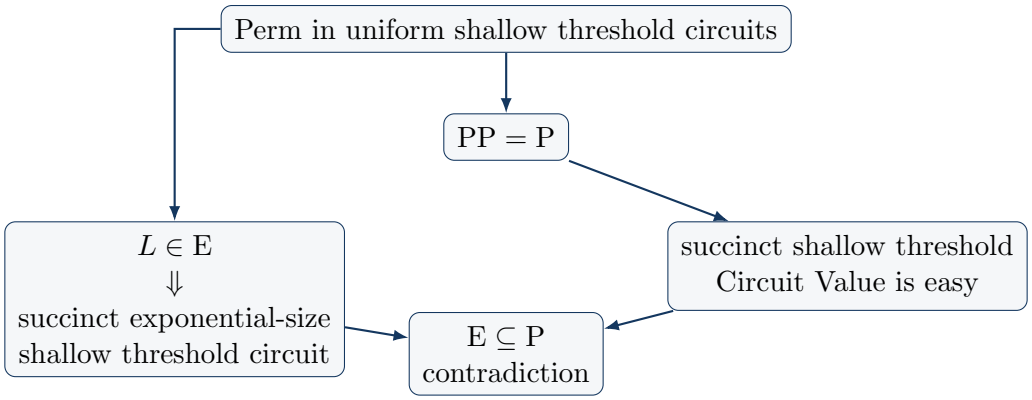
$$o(\log \log n).$$

Their proof is exactly the same indirect diagonalization, with the quantitative version of Lemma 3.2:

$$\text{succinct depth-}d \text{ threshold evaluation} \in \mathrm{DTIME}\big((\log s)^{2^{O(d)}}\big) \qquad \text{if PP} = \mathrm{P}.$$

For the exponentially large circuits obtained by scaling an E computation, $s = 2^{O(n)}$. Thus depth $d = o(\log n)$ after scaling yields a subexponential-time algorithm for every language in E, contradicting time hierarchy.

Earlier results optimize a different parameter. Allender–Gore proved much stronger *size* lower bounds for uniform constant-depth ACC^0 circuits computing the Permanent, while Allender proved strong lower bounds for uniform constant-depth threshold circuits. Koiran–Perifel sacrifice that size strength in exchange for allowing the depth to grow to $o(\log \log n)$.



8 What a course needs to know

For the constant-depth proof, the prerequisite list is unusually short.

Ingredient	**Needed fact**
Counting complexity	#P, PP, and #P-completeness of Permanent
Uniformity	DLOGTIME-uniform AC^0 and threshold circuits
Succinctness	exponentially large circuits can have polynomial-size local descriptions
Hierarchy	E $\not\subseteq$ P
Threshold circuits	one gate asks whether sufficiently many predecessors accept

No Toda theorem, switching lemma, Beigel–Tarui representation, or lower-bound measure is needed. In a course where counting complexity and uniformity have already been introduced, the proof itself is plausibly a half-lecture theorem.

References

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